

pktSNMP — Network Intelligence Platform

Real-time SNMP telemetry collection, visualization, and alerting for modern network infrastructure.

SNMP v1 / v2c / v3

All major protocol versions across local and remote devices

60-Second Alert Loop

12 built-in rule types — auto-fires and auto-resolves

Dual Storage

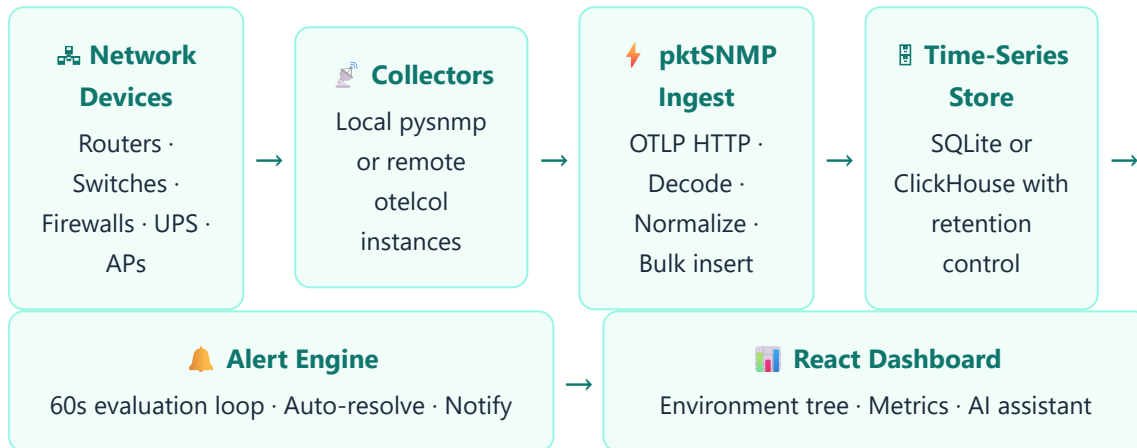
SQLite (zero-config) or ClickHouse (high-volume) — switchable live

AI + SAML

Natural-language queries + SAML 2.0 SSO with RBAC

End-to-End Data Flow

pktSNMP collects SNMP telemetry from every corner of your network and delivers it to operators in a unified real-time dashboard.



Collection Architecture

pktSNMP supports two complementary collection modes that can be used simultaneously:

Local Collector (built-in): The pysnmp poll engine and SNMP trap receiver (UDP 162) run in-process on the pktSNMP server. No additional infrastructure required. Configure via Settings → SNMP in the UI.

Remote otelcol Collectors: Existing OpenTelemetry Collector instances at distributed sites poll local devices and push OTLP HTTP data to pktSNMP. This approach extends visibility to remote locations without deploying additional server infrastructure at each site.

Key Features

🌐 Unified Device Visibility

Four-level hierarchy (Org → Group → Site → Device). Status dots roll up — one down device surfaces instantly at every level above it.

📊 Rich Metrics & Time-Series

Per-interface traffic (bps, packets/s), error/discard rates, link speed, and operational status — charted with configurable time ranges and CSV export.

🔔 Intelligent Alerting

12 rule evaluators: interface down, flapping, bandwidth saturation, error spikes, collector gaps, and more. Rules auto-resolve when conditions clear.

🤖 AI-Powered Assistant

Ask natural-language questions — "Which devices had errors in the last hour?" — and get instant answers sourced from your live telemetry.

🔑 Enterprise Auth

Role-based access (Admin / Analyst / Viewer) via local JWT or Okta SAML 2.0 SSO. Roles assigned at the IdP — no per-user config in pktSNMP required.

⚙️ Flexible Deployment

Single systemd service, zero external dependencies by default. Add remote otelcol at any site — no agent install on monitored devices.

Technology Stack

Layer	Technology
Backend	FastAPI + aiosqlite (Python 3.11+)
Frontend	React 18 + TypeScript + Tailwind CSS + Vite
Control-plane store	SQLite pktsnmp.db (users, devices, collectors, alert rules)
Time-series store	SQLite snmp_timeseries.db (default) or ClickHouse
Auth	JWT (15 min) + httpOnly refresh (7 days) + SAML 2.0

SNMP	pysnmp-lexstudio (v1/v2c/v3 traps and polling)
Collection	pysnmp local + otelcol-contrib remote (OTLP HTTP)
Service	systemd pktsnmp.service
Notifications	In-app · Email · Slack · PagerDuty · Webhook

User Roles & Access

Action	Admin	Analyst	Viewer
View dashboard, metrics, alerts	✓	✓	✓
Acknowledge alerts	✓	✓	—
Add / edit devices	✓	✓	—
Configure alert rules	✓	—	—
Manage users & settings	✓	—	—

Alert Rule Types

The alert engine evaluates these 12 rule types every 60 seconds:

Rule	Description
interface_down	Interface operational status changes to down
interface_flap	Interface state changes rapidly within a window
metric_threshold	A metric crosses a defined threshold
metric_spike	A metric spikes beyond a rolling baseline
error_rate	Interface error count exceeds threshold
discard_rate	Interface discard count exceeds threshold
high_error_ratio	Error ratio as a percentage of total packets
bandwidth_utilization	Bandwidth exceeds a % of interface speed
speed_change	Interface negotiated speed changes
collector_gap	Collector stops sending data (ingest gap detected)
device_unreachable	Device last_seen is stale beyond the threshold

trap_received	SNMP trap received from a registered (or unknown) device
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All rules auto-resolve when the condition clears — no manual intervention needed. Open events are marked resolved_at automatically, and device status returns to "up."